MILITARY SPECIFICATION

INTERCOMMUNICATION STATIONS

NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers LS-518( )/SIC and LS-519( )/SIC intercommunication stations for use in Naval shipboard announcing systems.

1.2 Classification. - The intercommunication stations shall be of the following types as specified (see 6. 2). The type designation for the intercommunication stations contains parentheses which are to be replaced by a letter indicating a particular make or model that differs from previous production. Nomenclature shall be requested as required by MIL-E-21981.

1.2.1 Type LS-518( )/SIC. - Type LS -518( )/SIC station shall be suitable for installation in protected locations and shall be capable of selecting 10 other stations in any combination of at least 2 stations re - gardless of type.

1.2.2 Type LS-519( )/SIC. - Type LS-519( )/SIC station shall be suitable for installation in protected locations and shall be capable of selecting 20 other stations in any combination of at least 2 stations regardless of type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27	- Transformers and Inductors (Audio, Power and Pulse).
MIL-C-62	- Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolytic, Polarized), General Specification For.
MIL-C -915	- Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-I-983	- Interior Communication Equipment; Naval Shipboard, Basic Design Requirements For.
MIL-C -2194	- Cables, Power, Electrical, Reduced Diameter Type, Naval Shipboard.
MIL-C-3965	- Capacitors, Fixed, Nonsolid, Electrolytic (Tantalum), Foil and Sintered Slug, General Specifications For.
MIL-L-12606	- Loudspeaker, Permanent Magnet (Unencased, 4-inch Diameter Cone, 2-Watt, Gunblast - and Immersion-Resistant), Type LS-445/V.
MIL-M-15071	- Manuals, Equipment and Systems.
MIL-T-15514	- Telephone, Sound Powered (Handsets, Head-Chest Sets, and Head-Transmitter Sets).
MIL-E-17362	- Electronic Repair Parts Requirements, Procedures for Provisioning Technical Documentation and Stock Numbering.
MIL-E-17555	- Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-I-17928	- Intercommunication Stations, Naval Shipboard Use.
MIL-A-20222	- Announcing-Equipment, Shipboard.
MIL-E-21981	- Electronics Type Designations, Identification Plates and Markings, Requireents For.
MIL-I-22560	- Intercommunication Stations, Hands-Free Operated Reply Reproducer, Naval Shipboard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD- 105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-200 - Electron Tubes, Selection and Use of.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.

DRAWINGS

BUREAU OF SHIPS

9000-S6502-74117 - Microphone-Plug and Receptacle
9000-S6502-74164 - Type IC/MPM-4 Microphone

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St. , New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - A preproduction sample shall be furnished for the tests specified in 4.2 (see 3.27 and 6. 3).

3.2 Description. - The intercommunication station, hereinafter referred to as the “equipment”, shall have selector switches for calling the required number of stations, a release light, a call light, a busy light, one press-to-talk switch, a microphone receptacle, a dimmer control, and one volume control switch. The equipment shall have a transistorized amplifier-speaker for voice amplification and trans - mission and shall be suitable for installation in protected locations.

3.3 General features. - The equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. (Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General requirements
Definitions
Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc -resistant materials
Toxic materials
wood
Aluminum
Iron and steel
Nonferrous material (except aluminum)
Zinc
Springs (material)
Other metals
Plastics

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

Ceramics
Impregnating, embedding and encapsulating compounds
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Parts-electrical-general
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Electrical tapes
Batteries
Dial illumination lamps
Switches
Indicator lights and lampholders
Fuses
Metallic rectifiers
Enclosure-degree of (see 3.9)
Enclosure -mounting
Minimum sheet metal thicknesses (watertight enclosures)
Cable entrance
Threaded devices
Rounded corners and edges
Welding
Temperature and humidity (special cold limit applies)
Shock, vibration and inclination (vital vibration test required)
Primary power supply circuits
Power supply tolerances (normal power supply, 115 volts, 60 cycle, single phase)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Terminal boards, connectors and terminals
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances
Drawings -general
Drawings -preliminary
Drawings-working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Interchangeability and standardization
Manuals (see 3. 30)
Repair parts (electronic) (see 3. 29)
Designation and marking (see 3. 28)
Nomenclature
Workmanship and general examination

3.4 Design.- The equipment shall be designed to provide naturalness and intelligibility of speech uncle r quiet conditions, penetrating ability of speech under noisy conditions, dependability of operation, and accessibility for servicing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3.5 Fuseholders. - Fuseholders shall be of a spring clip type and shall be silver plated.

3.6 Station selector switch. - The station selector switch shall be self-locking gang-type push-button assembly. For the type LS-518( )/SIC intercommunication station, the switch shall contain 10 switch pushbuttons and 1 release button. For the type LS-519( )/SIC intercommunication station the switch shall contain 20 switch pushbuttons and 1 ?? . release pushbuttons. When operated, all switch pushbuttons shall remain in the operated position, 'regardless of the sequence of operation, until the release button is depressed. Switch blades shall be of the bifurcated type. All switch contacts shall be of fine silver or better. Each station pushbutton shall be capable of withstanding 100,000 operations without failure when assembled in the equipment and connected for normal operation. Design of the switch shall be such that it will continue to perform its function after being subjected to the shock and vibration tests required by MIL-I-983.

3.7 Size and weight. - The equipment size and weight shall not exceed the following:

(a) LS-518( )/SIC

	Case	Overall (including mounting plate and handle grips)
Width	10-3/4 inches	11 inches
Depth	9 inches	10 inches
Height	10 inches	12-1/2 inches
Weight	42 pounds	

(b) LS-519( )/SIC

	Case	Overall (including mounting plate and handle grips)
Width	10-3/4 inches	11 inches
Depth	9 inches	10 inches
Height	10 inches	12-1/2 inches
Weight	48 pounds	

3.8 Microphone receptacle. - A microphone connector receptacle as shown on Drawing 9000-S6502-74117 shall be provided on the front panel of all units for use of microphones and handsets as specified in 3.24.6 and 3.25.7. Wiring to the microphone receptacle shall be as shown on figure 1.

3.9 Enclosure. - The equipment enclosure shall be designed for mounting on a bulkhead, in a panel, or on a desk. A removable mounting plate or similar means shall be provided to allow mounting of the unit on a bulkhead. With the mounting plate removed, the unit shall be suitable for mounting on a desk, using three or four mounting bolts on the bottom of the enclosure. The front panel including the amplifier and control circuits shall be removable from the enclosure. The enclosure shall be steel or aluminum. The enclosure shall be dripproof in accordance with MIL-I-983.

3.10 Front panel and case securing bolts. - Through bolting for the purpose of securing the front panel to the case and for securing the case in its mounted position will be permitted. All securing bolts shall have hexagonal heads with screwdriver slots. Sufficient space shall be allowed around bolt head surfaces to permit removal with a socket wrench.

3.11 Front panel surface. - The front panel shall be free of unnecessary projections. The depth of all operating controls in the nonoperated position shall be approximately the same distance from the front panel surface.

3.12 Handle grips. - Handle grips shall be provided on the f rent panel to facilitate withdrawal of the front panel from the case.

3.13 Electrical connections. -

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3. 13.1 Ship's wiring connections. - A terminal board shall be provided for connecting ship's wiring to the equipment, Sufficient terminals shall be provided on the terminal board to allow ship's wiring connections for all modes of operation.

3.13.2 Plug-in connectors. - Multi-contact plug and jack connectors shall be provided which will disconnect the unit from ship's wiring when it is removed from its case. The plug shall be mounted on the unit and the jack shall be mounted in the rear of the case such that the plug engages in the jack when the unit is placed in its case. The connectors shall employ a floating type of mount to assure positive alignment and positive connection.

3. 13.2.1 Alignment guides. - Alignment guides shall be provided to assure positive alignment between the plug and jack connectors. The guides shall support the unit during insertion and removal intervals and while in the case. The guides shall prevent any damage to the plug-in connectors or any part of the unit during insertion and removal of the unit from its case.

3.14 Drainage. - Drainage shall be provided for any panel recesses which otherwise would be susceptible to collection of water.

3.15 Control knobs. - The size and shape of the operating control knobs shall be suitable for the function for which they are intended.

3. 15.1 The position of all switch controls shall be clearly visible in the normal or operated position.

3. 15.2 Knobs and shafts shall have suitable flats or rectangular shapes on the connecting parts and shall be secured with a through roll-pin or other approved means.

3.16 Gaskets. - Gaskets shall be used, if necessary, for making dripproof seals between the case and front panel assembly. Gaskets shall be used for operating controls of the translating and rotary motion types.

3.17 Servicing. - With the unit removed from its case, all components shall be readily accessible for ease of replacement. When the unit is removed from its case for servicing, all component parts shall be protected from possible damage when placed on any of its sides.

3.18 Transformers. - Transformers shall be grade 5, class S, life expectancy X, as specified in MIL-T-27. Where special shielding problems exist, such as in input circuits of audio amplifiers, the use of encased type transformers employing high permeability alloy shields will be permitted.

3.19 Relays. - Relays if used, shall conform to MIL-I-983 and shall have removable dustproof covers.

3.20 Illumination and controls. - All illumination shall be red in accordance with MIL-I-983. Illumination lamps shall be located to permit direct replacement of each lamp individually from the exterior of the enclosure. Control and level of illumination shall be as follows:

(a) All signal indicators, busy, call, and release, shall be energized at normal brilliance. Control of signal indicators shall be accomplished by means of an "ON-OFF" switch. The switch shall be a part of the dimmer control for the station identification marker and shall be "OFF" in the minimum or fifth position of the dimmer controls. At normal brilliance, the level shall be such that the indications are visible at a distance of 18 inches when the equipment is mounted in areas exposed to sunlight.

(b) Normal brilliance of station identification markers shall be that level which is easily visible at a distance of 2 feet from the marker under average office lighting conditions with the equipment located at a desk or normal working position. Minimum brilliance shall be interpreted as the minimum level at which markings are easily readable at a distant e of 18 inches from the marker in a completely darkened room. This level shall be determined after a dark adaptation period of 10 minutes. A continuous or step type control shall be incorporated to provide an even increase or decrease of illumination between the limits specified herein. The "ON - OFF" switch required for the signal indicators shall be incorporated as part of this control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3.21 Signal indicator lamps. - The release indicator Lamp shall be located directly over the release switch. The call and bus y indicator lamp shall be individually located over the station selector switches. Signal indicator lamps shall be subminiature, omnidirectional, red plastic dome using a type T-1-3/4 lamp.

3.22 Station identification. - This shall be a durable, abrasive-resistant laminated plastic strip suitable for engraving. The laminations shall be opaque black and translucent white. Guide lines shall - be engraved on the marker to indicate the limits of the area to be engraved with the station designation. The engraved area shall be at a sufficient depth to expose the translucent white lamination. The marker shall be a single strip removable from the f rent of the equipment.

3.23 Rectifiers. - Silicon type rectifiers shall be used to supply high voltage for the amplifier. Suitable resistance shall be inserted in the high voltage circuit as a surge current limiter. Rectifiers' shall be selected from approved types listed in MIL-STD-200.

3.24 Operational features. - The operational features of the system shall be as specified in 3.24.1 through 3.24.7. Equipment furnished under this specification shall be designed to operate in a system which may include type LS-385/SIC or LS-386/SIC intercommunication stations manufactured in accordance with MIL-I-17928 or type LS-458/SIC intercommunication stations (hands-free operated reply reproducer) manufactured in accordance with MIL-I-22560. The schematic diagram for the LS-458/SIC intercommunication station is shown on figure 1 of this specification. Master stations referred to may be type LS-385/SIC, LS-386/SIC, or the type of intercommunication station to be furnished under this specification. Intercommunication stations furnished under this specification shall be electrically compatible with types LS-385/SIC, LS-386/SIC and LS-458/SIC such that any of the three types can be replaced by a unit furnished under this specification.

3.24.1 Communication between two master stations. -

3.24. 1.1 Normal operation to call. - Press station selector switch. If station selected is not busy (indicated by busy lamp) operate press-to-talk switch and talk into speaker.

3.24. 1.2 Normal operation to answer a call. - Operate press-to-talk switch and talk into speaker.

3.24. 1.3 Operation when calling a busy station. - When a station has received a busy indication, it shall not be possible to hear speech from the busy station.

3.24. 1.3.1 Busy override feature. - All units shall be provided with a busy override feature. By removal of jumper connections at ships terminal boards at each station, it shall be possibIe to rearrange the busy circuit so that a station receiving a busy indication can hear the communication of the busy station and talk to the busy station. Under this condition the station calling the bus y station may superimpose speech signals on those of the busy station.

3.24. 1.3.2 Operation with busy override. - If busy indication is received (see 3.24.1. 1) operate the press-to-talk switch. This will cause the busy light to go out and speech from the busy station or station to whom the busy station is talking to be heard. To talk to the busy station, operate the press-to-talk switch and speak into the loudspeaker.

3.24.2 Parallel operation of two master stations. - When two units are connected for parallel operation they shall be the same as for normal single operation with the following exceptions:

(a) Incoming speech will be heard at both units.
(b) Either station may answer a call but not both. (Inadvertent answering by both units simultaneously will not cut off transmission of speech to the calling unit.)
(c) Operation of volume control from either unit will control volume of both units.
(d) Outgoing speech from one paralleled unit will not be heard at the unit to which it is paralleled.
(e) Operation of a station selector at either of the paralleled units in addition to lighting the call light at the called station will cause the call light in the unit to which it is paralleled to light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3.24.3 Communication between a master station and an LS-458/SIC.- When used as a master station the equipment shall be capable of working with an LS-458/SIC to provide the following operational features:

3.24.3.1 LS-458/SIC calling equipment (master) station. - To call the equipment station the LS-458/SIC operator shall operate the station selector switch. If the master station selected is not busy (indicated by busy lamp) the LS-458/SIC operator shall operate the press-to-talk switch and talk into the speaker.

3.24.3.2 Master station calling LS-458/SIC. - When called by the equipment station the LS-458/SIC shall operate as follows to reply to the call:

(a) Normal operation to answer a call. - With "Remote-Normal" switch on LS-458/SIC station in "Normal" position operate press-to-talk switch and talk into speaker.
(b) Hands free operation. - With "Remote-Normal" switch on LS-458/SIC station in "Remote" position, operation of station selector switch on equipment (master) station shall put the LS-458/SIC in "hands-free talk" condition. Operation of press-to-talk switch on equipment station shall cause the LS-458/SIC station to be in the "Listen" condition. When the master station operator depresses the release switch both stations shall go to standby condition.

3.24.4 Hands free operation of equipment station. - When the intercommunication stations furnished under this specification are used with LS-385/SIC and LS-386/SIC intercommunication stations to make up an intercommunicate ion system, any of the stations furnished under this specification shall be capable of replying hands free when called by another station in the system. Operation shall be as follows:

3.24.4.1 Operation of the calling station to originate a call. -

(a) Operate station selector switch to select station to be called.
(b) Operate press-to-talk switch to talk to called station. (Operation of calling station press-to-talk switch shall cause the called station to be in a "listen" condition.)
(c) Release the press-to-talk switch to listen to reply from the called station. (Releasing the calling station press -to-talk switch shall cause the called station to be in the "talk" condition.)
(d) Returning station selector switch of calling station to OFF or release position shall return both calling and called stations to standby condition.

3.24.4.2 Operation of called station to reply to an incoming call. - The press-to-talk switch of the equipment station shall be set to the "hands free" position for this type of operation. To reply to an incoming call the only action required of the called station operator is to talk into the loudspeaker of his station. (All switching for "talk" and "listen" are controlled by the calling station.)

3.24.5 Operation of equipment with remote reproducer. - This system shall consist of one equipment station, one remote reproducer (LS-306A/SIC loudspeaker manufactured in accordance with performance requirements of MIL-A-20222 for loudspeaker IC/SAG), and a normally closed single pole switch located near the reproducer. Circuits shall be provided for the following operations:

(a) Operation of remote reproducer switch will connect remote reproducer to the input of the equipment amplifier and connect the equipment reproducer to the output of the equipment amplifier. The remote reproducer can then transmit speech.
(b) Operation of station selector switch on the equipment will connect the remote reproducer to input of the equipment amplifier and connect the equipment reproducer to output of the equipment amplifier. Speech can then be transmitted from the remote reproducer without operation of the remote reproducer switch.
(c) With equipment selector switch operated, this station can transmit speech to remote reproducer by operation of press-to-talk switch. Remote reproducer can reply with hands - free operation when the equipment "press-to-talk" switch is in restored position.

3.24.6 Operation with handset or microphone. - Operation with handset or microphone shall be identical to that when the internal loudspeaker is used as a microphone, except that press-to-talk functions will be performed remotely by the press-to-talk switch in the handset or microphone.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3.24.7 Operations directly associated with the station selector switch.-

3.24.7.1 Release lamp. - Operation of the station selector switch in selecting a station to be called sha11 illuminate the release lamp of the calling station. This light shall serve to indicate that a station selector button is depressed at the caIling unit and should be released at the end of a conversation. Depressing the release button to restore the station to standby shall extinguish the release lamp.

3.24.7.2 Call lamp. - Operating the station selector switch of a calling station to originate a call shall cause the call indicator lamp of the called station to be illuminated providing the called station is not bus y.

3.24.7.3 Busy indicator circuit. - When the calling station selects a station which is busy the busy indicator lamp of the calling station shall light and the calling station shall be prevented from transmitting messages to or receiving messages from the called station. Restoring the selector switch to normal or standby shall turn off the busy light and restore the station to normal receiving condition.

3.24.7.4 Privacy. - Operation of the station selector switch shall cause the selected station to become busy to all other stations in the system. Operation of this switch shall also cause the selecting station to become busy to all other stations in the system. Once communication has been established by operation of station selector switches, no other nonselected units shall be able to listen or to break in and talk to any units engaged in conversation except under conditions specified in 3.24.1.3.2.

3.25 Electrical features. -

3.25.1 Circuit design. - The circuit design shall perform the functions specified in 3.24 through 3.24.7.

3.25.2 Terminal markings. - Terminal markings shall be in accordance with MIL-I-17928.

3.25.3 Interconnecting cables. - Type TTHFWA cable conforming to MIL-C-915 will be employed for interconnection of audio and control circuits when installed. Reduced diameter cable conforming to MIL-C-2194 shall be employed for power and signal circuit common. A maximum length of 1500 feet shall be used for calculating line impedance of audio and signal circuits.

3.25.4 Volume control. - The volume control shall provide a maximum attenuation of 30 + 1 decibels (db) in five steps of 6 ± 1 db. No "OFF" position shall be incorporated in the control.

3.25.5 Rotary and toggle switches. - The handles or keys of rotary type switches shall be provided with mechanical stops to prevent excessive strain on the switch handle and shaft, and to prevent rotation of the fixed portion of the switch. All switches shall, after 100,000 operations, when energized at their rated loads be capable of performing both mechanical and electrical functions for which they are intended.

3.25.6 Press-to-talk switch. - The press-to-talk switch shall be a three position switch which performs the necessary switching required for operation specified in 3.24. The three positions of the switch reading from the front panel in a clockwise direction shall be "hands free", "normal" and "press-to-talk". The switch shall be spring loaded such that when released it will return from the "press-to-talk" position to the "normal" position. When operated from the "normal" to the "hands free" position the switch shall lock in the "hands free" position until it is again operated to the "normal" position. The "hands free" position of the switch shall be used to set up the equipment for the operation specified in 3.24.4.

3.25.7 Microphone and telephone handset. - The equipment shall be designed to permit the use of a type IC/MPM-4 microphone or a type IC/MPH telephone handset which may be connected to the unit by means of the microphone receptacle specified in 3.8. The design specifications for the type IC/MPM-4 microphone and for the type IC/MPH telephone handset (with the exception of the receiver element) are shown on Drawing 9000-S6502-74164. The design specifications for the telephone handset receiver element is MIL-T-15514. Operation with *a* microphone or handset shall be as specified in 3.24.6.

3.25.8 Loudspeakers. - Loudspeakers shall be type LS-445/U in accordance with MIL-L-12606.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3. 25.9 Transformer polarities Identical instantaneous alternating current (a. c.) polarities for all power transformers shall be obtained at physically identical transformer terminals on all stations when all stations are energized from the same source.

3.26 Amplifier. -

3.26.1 Transistors and metallic rectifiers. - Transistors shall be used in the design of all amplifiers. Transistors shall be types chosen from approved types listed in MIL-STD-701. Power rectifiers shall be chosen from approved types listed in MIL-STD-200.

3.26.2 Printed wiring. - Printed wiring if used, shall be in accordance with MIL-STD-275 and type C lead terminations shall be employed.

3.26.3 Protection of signal voltage winding of power transformer. - Fuses shall be incorporated in the signal voltage circuit to prevent damage to the windings of the transformer due to defective signal voltage circuits.

3.26.3.1 Fuses shall be inserted in each lead of the primary power supply in the unit.

3.26.4 Capacitors. - Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in the amplifier assemblies.

3.26.4.1 Electrolytic capacitors in accordance with MIL-C-62 may be used in power supply filter circuits only.

3.26.5 Personnel protection. - In the case of voltages in excess of 240 volts, a red warning plate shall be mounted in a conspicuous location in accordance with MIL-I-983.

3.26.6 Heat dissipation. - Components which generate heat shall be mounted to allow maximum heat dissipation through the external case.

3.26.7 Rated power output. - Rated power output shall be 3 watts at a maximum distortion of 10 percent.

3.26.8 Output voltage. - The output voltage when delivering rated power shall be 70 volts ± 4 volts root mean square (r. m.s.).

3.26.9 Amplifier gain. - The amplifier gain shall be sufficient to deliver rated output power at rated output voltage with a total harmonic distortion of 10 percent with a 1000 cycle sine wave voltage equivalent in value to the voltage generated by the loudspeaker, when used as a microphone under the test conditions specified in 4. 6.2 applied to the input terminals of the amplifier. The amplifier gain shall be such that the amplifier will also deliver rated output power at rated output voltage when tested with a type IC/MPM-4 microphone or equivalent which will be capable of delivering 8.7 millivolt (rev.) into a resistive load of 150 ohms.

3.26.10 Output voltage regulation. - With a steady state sine wave input voltage, of the same value used to determine amplifier gain, applied to the input terminals of the amplifier, a variation of the amplifier output load from full load to no load shall not cause the output voltage to exceed 99 volts r. m.s. at any frequency between 200 and 8000 cycles per second (c. p.s.).

3.26.11 Stability. - The amplifier shall show no evidences of instability of oscillation under any condition of operation.

3.26.12 Amplifier noise and microphonics. - The amplifier noise and microphonics shall not exced the following values when tested as specified in 4.6.7.

(a) Component stationary. - Maximum measured voltage shall not exceed 0.22 volts r. m.s. (50 db below rated output voltage of 70 volts).
(b) Component vibrating. - Maximum measured voltage shall not exceed 0.31 volts r. m.s. (47 db below rated output voltage of 70 volts).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

3.26.13 Overall frequency response. - The overall frequency response using the loudspeaker as a microphone shall fall within the limits of the overall frequency response curves shown on figure 2 when tested as specified in 4.6.4 and 4.6.5.

3.26.14 Minimum sound pressure. - The loudspeaker, when mounted in the front panel, shall be capable of producing the following minimum sound pressure at a distance of 2 feet in front of and on axis with the loudspeaker when measured as specified in 4.6.3.

Warble band c.p.s.	200 to 320	320 to 500	500 to 800	800 to 1250	1250 to 2000	2000 to 3200	3200 to 5000
Output pressure db 1/	85	95	99	99	99	99	99

1/ Above 0.000200 dynes per Cm^2.

3.26.15 Amplifier endurance. - After being tested as specified in 4.6.6, the equipment shall be capable of normal operation in every respect. There shall be no visible damage to any electrical or mechanical component of the equipment.

3.26.16 Signaling circuits. - The busy, and call signals shall not fail to indicate nor shall speech transmission be audible at a station receiving a busy signal when tested as specified in 4.6.8.

3.26.17 Amplifier output circuits. - The paralleling of amplifier output circuits which can occur if two stations call a third station simultaneously shall not cause a loss of signal greater than 2 db, nor shall the distortion exceed the maximum allowable distortion.

3.27 Test fixture. - A test fixture for testing type LS-518( )/SIC and type LS-519( )/SIC stations shall be included in the maintenance e parts. When a preproduction model of the equipment covered by this specification is submitted to a Government laboratory for testing a test fixture shall be submitted with the model for use in testing. The arrangement of the test fixture and method of fastening to the station unit shall be such that the test fixture can be easily attached to the back of the unit when the unit is removed from its enclosure. A portable power cable shall be furnished with the test fixture. The test fixture shall be designed to permit testing of the call lamp circuit, release lamp circuit, busy lamp circuit, polarity, amplifier and reproducer operation, audio circuit through each station selector switch, hands free reply circuit operation, and other tests that may be applicable. There shall be provision for applying a dummy load on the amplifier for servicing.

3.28 Designation and marking. - An instruction plate containing abridged operating instructions shall be provided on the outside of the enclosure. Schematic circuit diagrams shall be provided and mounted inside the enclosure so as to be usable in servicing the unit. The circuit diagram plates may be of aluminum or plastic material. Designation plates shall be provided for all control knobs. These plates shall be located adjacent to the knobs and shall clearly indicate the function and direction of rotation of the knobs.

3.29 Equipment repair parts. - Equipment repair parts shall be in accordance with MIL-E -17362.

3.30 Manuals. - The manual shall be type I in accordance with MIL-M-15071. The manual shall cover the equipment specified herein, and shall include microphone and handset operating techniques. The manual shall contain preliminary drawings, working drawings, and an illustrative schematic showing the interconnection of stations and a sufficient portion of the internal wiring of each station to enable a reader to determine the operation of the switches and relays and to service and install the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests specified in table I, which shall be performed in general in the order listed.

Table I- Preproduction inspection.

Inspection	Requirement paragraph	Test reference
General examination	3.3	MIL-I-983
Operating	3.24	4.6.1
Supply line voltage and frequency variation	3.3	MIL-I-983
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Temperature and humidity	3.3	MIL-I-983
Inclination	3.3	MIL-I-983
Insulation resistance	3.3	MIL-I-983
Amplifier gain	3.26.9	4.6.2
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Amplifier endurance	3.26.15	4.6.6
Signaling circuits	3.26.16	4.6.8
Enclosure	3.3 and 3.9	MIL-I-983
Shock, vibration and inclination	3.3	MIL-I-983

4.3 Sampling for quality conformance inspection. -

4.3.1 Lot. - All equipment presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.3.2 Sampling for group A tests. - Sample equipments shall be selected in accordance with MIL-STD-105 at inspection level III for the tests specified in 4.4. The acceptance quality level (AQL) shall be 1.5 percent.

4.3.3 Sampling for group B tests. - Sample equipments shall be selected in accordance with the table H for the tests specified in 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

Table II- Sampling for group B tests.

Lot size number of equipment	Sample size number to be tested	Test failures	
		Number allowed	Reject the lot on
Upto15	3	0	i
16 to 50	5	0	1
51 to 90	8	0	1
91 to 280	13	1	2
281 to 500	20	2	3
501 to 1200	32	3	4

4.3.4 Sampling for group C tests. - Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected and subjected to the group C tests listed in 4.4.

4.4 Quality conformance inspection. - The sample equipments selected in accordance with 4.3 shall be subjected to the examination and tests specified in table 111. Tests shall be performed, in general, in the order listed.

Table HI-Quality conformance inspection.

Inspection	Requirement paragraph	Test reference
Group A		
General examination	3.3	MIL-I-983
Operating	3.24	4.6.1
Amplifier gain	3. 26.9	4.6.2
Group B		
Supply line voltage and frequency variation	3.3	MIL-I-983
Amplifier noise and microphonics	3.26.12	4.6.7
Overall frequency response	3.26.13	4.6.4 and 4.6.5
Minimum sound pressure	3.26.14	4.6.3
Signaling circuits	3.26.16	4.6.8
Group C		
Shielding and radio frequency noise reduction	3.3	MIL-I-983
Amplifier endurance	3.26.15	4.6.6
Temperature and humidity	3.3	MIL-I-983
Shock, vibration and inclination	3.3	MIL-I-983
Enclosure	3.3 and 3.9	MIL-I-983

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

4.5 Rejection. - The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A, B or C test shall be counted as a defect, and the equipment shall not be offered for delivery.

4.6 Test procedures. -

4. 6.1 Operating test. - The operating test shall consist of a voice transmission test for each intercommunication station with a complete operational test using a type LS-385/SIC or LS-386/SIC unit and an LS-458/SIC unit.

4.6.2 Amplifier gain. - The sound source for the test shall be the test laboratory speech generator (artificial voice) adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at a frequency of 1000 c. p.s. under free field conditions at a point of axis and 18 inches in front of the ring on the speech generator. The loudspeaker to be tested shall be mounted in its front panel and placed 18 inches in front of and on axis of, and facing the ring on the speech generator. The loudspeaker shall be connected to a resistive load corresponding in value to the normal amplifier input impedance. The voltage developed across the resistor shall be measured to determine conformance with 3.26.9.

4.6.3 Sound output pressure test. - Three type LS-518/SIC or LS-519/SIC intercommunication stations shall be connected together for normal operation. One of the stations shall be placed with its reproducer at a point 18 inches in front of and on axis and facing the ring of the artificial voice. A sound source from the artificial voice shall be adjusted so as to deliver a sound pressure of 4 dynes per cm^2 at the warble band frequencies specified in 3.26.14 under free field conditions at a point on axis and 18 inches in front of the ring on the artificial voice. The station under test shall have the selector key for the two other stations depressed and the press-to-talk switch shall be operated. The sound output pressures at the two other stations (with volume control in maximum position) shall be measured. Both stations being transmitted to shall be acoustically insulated from each other.

4.6.4 Overall frequency response test using loudspeaker as a microphone. - Conditions of overall frequency response test shall be as specified in 4.6.3 except that, in lieu of the warble frequencies specified in 3.26.14, a constant sound pressure at a frequency which shall be varied continuously from 200 to 8000 c. p.s. shall be employed. The overall frequency response shall lie within the limits shown on figure 2.

4.6.5 Overall frequency response using a type IC/MPM-4 microphone. - Test conditions shall be as specified in 4. 6.4 except that the sound source pressure shall be 100 dynes per cm2 and the distance from the microphone to the artificial mouth shall be 1/4 inch.

4.6.6 Amplifier endurance test. - A resistive load of 1667 ohms shall be connected across the normal output terminals of a station. A 1000-cycle sine wave signal of such value as to produce rated output at rated output voltage shall be applied to the microphone input terminals. The station shall be energized and the press-to-talk switch shall be depressed for a period of 7 seconds and then released for a period of 15 seconds. Immediately upon completion of 20,000 operations the station shall be connected for normal operation. There shall be no noticeable change in the quality of speech transmitted, nor shall the output decrease greater than -2 db from the output measured at the beginning of the test. Amplifier endurance tests shall be conducted in the hands-free mode, at rated load, for a period of 24 hours.

4.6.7 Amplifier noise and microphonics. - With a 150-ohm resistor connected to the microphone input terminals, and full power applied to the amplifier, the output voltage across a resistive load (of such value as to draw rated output power at 70 volts) shall not exceed the following values when measured with an unweighed full range voltmeter. The measurement shall be made under two conditions of test, once while the component is held fixed and again while the component is under vibration in accordance with 3.26.12.

(a) Component stationary. - Maximum measured voltage shall be O. 22 volt r. m.s. (50 db below rated output voltage of 70 volts).
(b) Component under vibration. - Maximum measured voltage shall be O. 31 volt r. m.s. (47 db below rated output voltage of 70 volts).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

4.6.8 Signaling circuits .- A combination of three intercommunicating units shall be connected for normal operation. Cable used to connect the equipment shall be equivalent to the cables specified in 3.25.3. The equipment shall be tested in the following sequence:

(a) At station Number 1, select station Numbers 2 and 3. The call lights at station Numbers 2 and 3 should light. Repeat this test ten times.
(b) At station Number 1, operate selector switch to select station Number 2. At station Number 3. operate selector switches to select stations Numbers 1 and 2. consecutively. The busy light at station Number 3 should light after each selection. This test shall be repeated ten times. A message shall be transmitted at station Number 1 during the fifth and tenth test cycles. A listening test shall be made at station Number 3 during the fifth and tenth test cycles. The transmission being made at station Number 1 shall not be audible at station Number 3.
(c) The test specified in 4.6. 8(b) shall be repeated until each station in the system has served as the calling station (station Number 1) and as the busy testing station (station Number 3).
(d) Tests shall be conducted for compliance with 3.26.17 by operating station selector switches on two equipment stations, simultaneously completing a call to a type LS-385/SIC or LS-386/SIC station.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5. 1.1 Intercommunication stations. -

5. 1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation, and may conform to the supplier's commercial practice when such meets these requirements.

5.1. 1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5. 1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number. manufacturer's name and destination.

5. 1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged by level A; packed by level C, and marked by levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procurnig activity (see 6. 2).

(5. 2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made:

5. 2.1.1 Preservation and packaging, packing and marking. - The equipment and accessories, repair parts and technical publications shall be preserved and packaged by levels A or C; packed by levels A or B as specified, and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. - The equipment covered by this specification is intended for intercommunication aboard Naval ships by means of voice amplification and station selection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1. 2).
(c) Preservation and packaging, packing and marking required if other than as specified in 5.1 (see 5.2).

6.3 Preproduction. - Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - SH
(Project 5835-N099Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

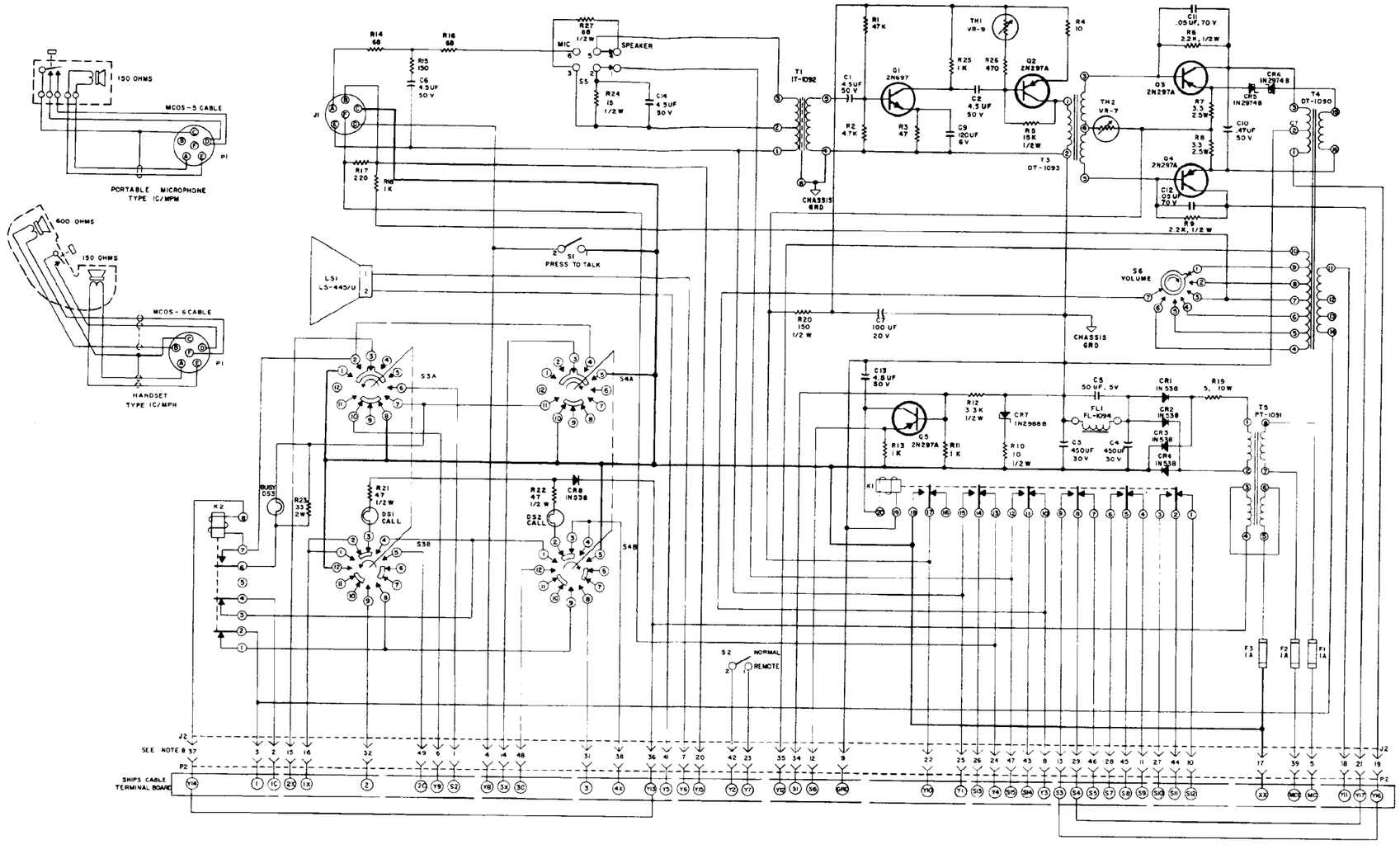


Figure 1 - Schematic diagram for Type LS-458/SIC Intercommunication Station.

NOTES

1. LINE VOLTAGE: 115 V, 60 CPS SINGLE PHASE
2. POLARITY: WITH MC CONNECTED TO XX, VOLTAGE FROM MCC TO Y13 IS GREATER THAN LINE VOLTAGE
3. INPUT POWER
   A. STANDBY: 9.5 WATTS
   B. OPERATING: 20 WATTS (.17 AMPERES)
4. OUTPUT POWER: 3.0 WATTS, 70 VOLT LINE DISTORTION 10% MAX.
5. INPUT IMPEDANCE AT MICROPHONE JACK, TERMINALS A & E, 150 OHMS
6. OUTPUT IMPEDANCE: 1667 OHMS
7. TERMINAL NUMBERS ON RELAYS K1 & K2 REFER TO PIN NUMBERS ON THEIR SOCKETS
8. DESIGNATIONS BETWEEN J2 & P2 REFER TO TERMINALS ON 50 CONTACT CONNECTORS, WHICH CONNECT THE EQUIPMENT TO THE SHIP'S WIRING TERMINAL BOARD
9. ALL RESISTORS ARE 1/4 WATT, EXCEPT WHERE OTHERWISE STATED
10. 66 DB 1000 CPS GAIN FROM TERMINALS 1 & 3 OF T1 TO TERMINAL 11 & 14 OF T4
11. TO REDUCE HANDSET RECEIVER VOLUME, PLACE RESISTOR FROM Y4 TO Y15 AS SHOWN IN TABLE

VOLUME REDUCTION, DB	RESISTANCE OHMS
3	150 , 1/2 WATT
6	68 , 1/2 WATT
9	33 , 1/2 WATT

12. ALL VOLTAGES ARE DC MEASURED WITH A 20000 OHM/VOLT, V.O.M. TO GROUND (-)

TRANSISTOR	TRANSISTOR ELECTRODE		
	E	B	C
Q1	0.5	1.1	1.6
Q2	16	16	0.4
Q3	19.5	20	0.2
Q4	19.5	20	0.2
Q5	20	12	12

13 TRANSFORMER TABLES

T1

WINDING NO'S	KC IMPEDANCE NOM	NO OF TURNS	WIRE GA B&S
1-2	8	100	32
1-3	150	400	32
4-5	250	500	35

T3

WINDING NO'S	KC IMPEDANCE NOM	NO OF TURNS	WIRE GA B&S
1-2	250	802	34
3-4-5	80	142 CT	32

T4 (AS OUTPUT TRANSFORMER)

WINDING NO'S	KC IMPEDANCE NOM	NO OF TURNS	WIRE GA B&S
1-2-3	175	144 CT	26
11-12-14	1666	440 CT	34
13-14	100	110	34
15-16	20	54	28

T4 (AS LINE TRANSFORMER)

WINDING NO'S	KC IMPEDANCE NOM	NO OF TURNS	WIRE GA B&S
11-14	3300	440	34
10-4	8	32	25
10-5		16	
10-6		8	
10-7		4	
10-8		2	
10-9		1	

T5

WINDING NO'S	VOLTAGE	NO OF TURNS	WIRE GA B&S
1-2	22	210	27
3-4	10	88	27
5-6	6	53	27
7-8	115	890	31

FL1

WINDING NO'S		NO OF TURNS	WIRE GA B&S
1-2		198	23

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

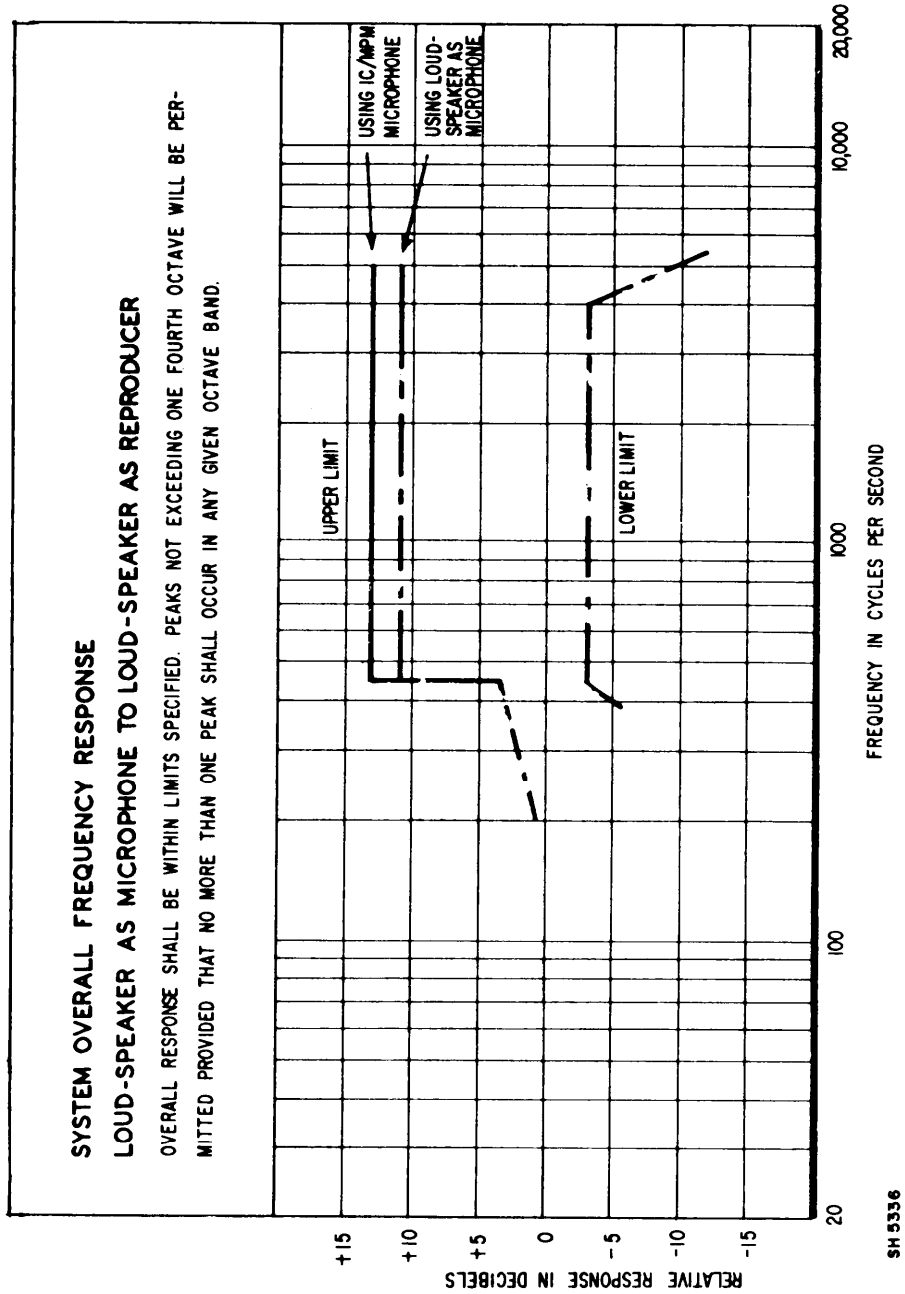


Figure 2 - System overall frequency response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Sea System Command (SEA 3112)
Department of the Navy
Washington, DC 20362

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T18:56Z
Check the source to verify that this is the current version before use.